

Modeling cyclic voltammetry and electrochemical impedance spectroscopy measurements of PEDOT:PSS layers with finite density of states

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ABSTRACT

This paper examines the impact of a finite density of states on hole transport in conjugated polymers, with a focus on poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS). To this end, a two-phase drift-diffusion model of PEDOT:PSS is extended to include a Fermi–Dirac, near-equilibrium occupation of the available hole states and a generalized Einstein relation between hole diffusivity and mobility in the PEDOT phase. A dedicated procedure is developed to impose the boundary conditions and to strengthen the convergence of the model equations. Numerical solutions of the extended model are compared to the results that neglect the above-mentioned aspects. We identify those conditions where the limited DoS affects the cyclic voltammetry and the electrochemical impedance spectroscopy response of a circular PEDOT:PSS-coated electrode in contact with an electrolyte. The results provide new insights into the electronic and ionic behavior of these electrodes and can be useful for preventing the inaccurate extraction of PEDOT:PSS properties or even the misinterpretation of cyclic voltammetry and electrochemical impedance spectroscopy experiments.

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I. INTRODUCTION

Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is a conjugated polymer/polyelectrolyte blend widely used in a variety of fields, including electrochemistry,^{1,2} flexible electronics,³ biosensing,⁴ and photovoltaics.⁵ In recent years, PEDOT:PSS has emerged among other conjugated polymers for its easy processing, controllable aqueous content,⁶ transparency, and chemical stability.³ The material is composed of a conductive polymer (PEDOT) and a polyelectrolyte counterion (PSS) that electrically stabilizes the material and improves its solubility in aqueous media.

Much endeavor has been devoted to the understanding of the microscopic mechanisms that regulate the electrical transport and modify the optical and thermal properties of PEDOT:PSS in contact with an electrolyte. In addition to the theoretical modeling presented in,^{7–9} extensive simulations—mainly based on mean-field, continuum, drift-diffusion models of charge transport—were carried out in Refs. 10 and 11 to identify which sets of equations best represent the material’s physical behavior.

The monomers composing a single polymeric chain of PEDOT can be electrochemically doped and host a hole.¹² These holes occupy localized energy states below the highest occupied molecular orbital (HOMO). Being PEDOT a soft material, with high electron–phonon coupling, the charged monomer quickly relaxes to a new geometrical conformation with lower energy, thus forming a polaron¹³ that introduces a new energy level in the bandgap. The hole density of states (DoS), which reflects the randomness of the polymer structure, is usually modeled using the Gaussian Disorder Model (GDM) originally introduced by Bässler in Ref. 7 where the Gaussian energy distribution represents the localized energy states below the HOMO. More recently, the Gaussian DoS has also been used to reproduce experimental electric and thermoelectric properties such as electronic conductivity and Seebeck coefficient in PEDOT-based conjugated polymers^{14,15} and I/V characteristics in PEDOT-based organic/inorganic devices.¹⁶

To maintain focus on the essentials of the physical problem, and in line with Refs. 11, 17, and 7, in this work, the hole DoS in the PEDOT phase is modeled with a Gaussian shape,⁷ while the

additional polaronic and bipolaronic energy states (generated in the bandgap by electrochemical doping as discussed above) are neglected in the following. In this context, the molar concentration of holes in PEDOT at equilibrium can be expressed as

$$p = p_t G(\eta, E_{\text{DoS}}, \sigma) \quad (1)$$

$$= p_t \int_{-\infty}^{\infty} \text{DoOS}(E) dE \quad (2)$$

$$= p_t \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(E-E_{\text{DoS}})^2}{2\sigma^2}} \cdot [1 - f(E, \eta)] dE, \quad (3)$$

where p_t (mol/m³) is the total molar concentration of hole sites, η is the chemical potential of holes, $f(E, \eta) = 1/[1 + \exp((E - \eta)/k_B T)]$ is the Fermi–Dirac (FD) occupation function, and E_{DoS} and σ are the mean and standard deviation of the Gaussian DoS, respectively, $\text{DoOS}(E)$ (units of J⁻¹) is the energy density of occupied states, T is the absolute temperature, and k_B is the Boltzmann constant. In the following, the normalized hole chemical potential $\zeta = (\eta - E_{\text{DoS}})/k_B T$ will be used as well.

If there are enough available states [i.e., $\text{DoS}(E)$ is large], and the occupation probability $f(E)$ is small (non-degenerate condition), then Pauli's exclusion principle can be ignored and the Maxwell–Boltzmann (MB) distribution can be safely used as an excellent approximation of the Fermi–Dirac occupation probability at equilibrium. Conversely, if $f(E)$ gets close to 1 (degenerate condition), then the Pauli exclusion principle should be taken into account. Nonetheless, to the best of our knowledge, Pauli exclusion and Fermi–Dirac (FD) statistics have rarely been considered for PEDOT:PSS, except in pioneering works from Bitton *et al.*, which focuses on models for organic electrochemical transistors,¹⁰ and Keene *et al.* which considers the generalized Einstein relation in the hole transport.¹⁸ However, these works do not explore the effects of Pauli exclusion on common material characterization techniques such as cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS), which is instead the main object of our investigation.

The integral in Eq. (3) can be worked out analytically if $f(E, \eta)$ is approximated with the MB expression, whereas it must be solved numerically if the FD occupation is adopted for accurate calculations under degenerate conditions. In these conditions, p/p_t , i.e., the DoOS integral in Eq. (3), differs considerably when considering either MB or FD statistics, as shown in Fig. 1. Therefore, a careful scrutiny of carrier degeneracy and the effects of Pauli exclusion is necessary.

In this work, we analyze the importance of conjugated polymer degeneracy (i.e., when the DoS is limited and the use of FD statistics becomes mandatory), particularly focusing on PEDOT:PSS.

The work proceeds as follows: Section II describes a finite elements (FEM) implementation of an extended model for hole transport in degenerate conjugated polymers, as well as the parameters used in the simulations, putting emphasis on how the finite DoS and the hole degeneracy are treated. Sections III and IV analyze the impacts of low DoS, degeneracy, generalized Einstein

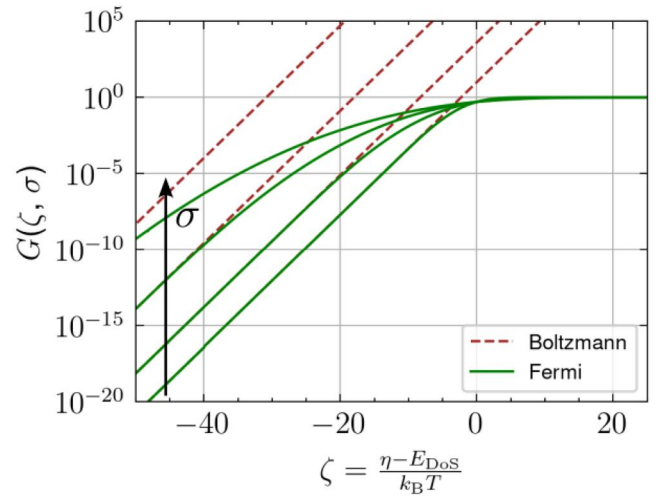


FIG. 1. Integral of the DoS (i.e., $G(\eta, E_{\text{DoS}}, \sigma)$) as a function of ζ , for MB and FD occupation probability in a typical range of ζ values ($V_{\text{app}} \in [-1 \text{ V}, 0.5 \text{ V}]$) and $\sigma \in [0.055, 0.105, 0.155, 0.205] \text{ eV}$. The typical σ of PEDOT:PSS is $\approx 0.105 \text{ eV}$.¹⁹

diffusivity, and degraded hole transport parameters on the CV and the EIS curves. Conclusions are drawn in Sec. V.

II. MODEL AND PARAMETERS

Following the work of Tybrandt *et al.*,¹¹ and with reference to Fig. 2, we implemented in COMSOL Multiphysics® two coupled sets of Poisson–Nernst–Planck (PNP) equations: the first one is

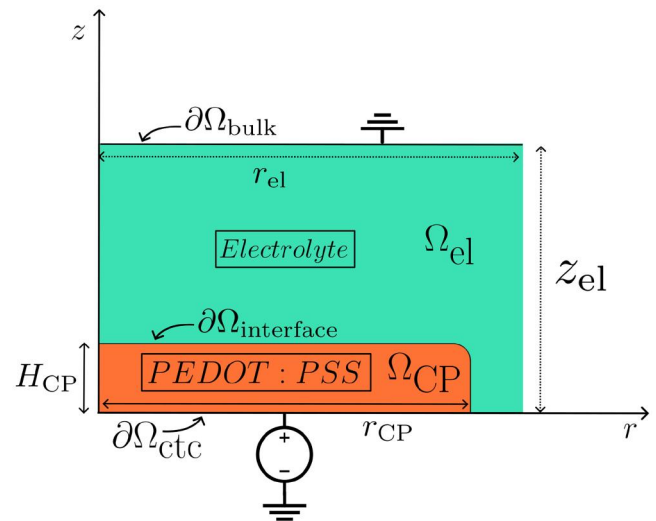


FIG. 2. 2D, cylindrical, z-axis-symmetric simulation domain of a conductive polymer immersed in an electrolyte. The dimensions are given in Table I.

TABLE I. Default model parameters used in the simulations.

H_{CP}	650 nm	T	300 K
r_{CP}	$3.5 \mu\text{m}$	$[KCl]$	154 mM
$r_{el} = z_{el}$	$20 \mu\text{m}$	D_K	$19.6 \times 10^{-10} \text{ m}^2/\text{s}$ (Ref. 21)
C_v	59 F/cm^3	D_{Cl}	$20.3 \times 10^{-10} \text{ m}^2/\text{s}$ (Ref. 21)
$\epsilon_{r,w}$	80	R_{CP}	2 M
$\epsilon_{r,PSS}$	80	E_{DOS}	4.1 eV
$\epsilon_{r,PEDOT}$	40	σ	0.105 eV (Ref. 19)
D_{p0}	$2.8 \times 10^{-9} \text{ m}^2/\text{s}$ (Ref. 22)	μ_{p0}	$1.08 \times 10^{-7} \text{ m}^2/\text{V s}$ (Ref. 22)
Ψ_{Au}	4.5 eV	Ψ_{AgCl}	4.7 eV

solved in the PEDOT:PSS domain (Ω_{CP}) and tracks the holes in the PEDOT phase; the second one covers the PEDOT:PSS and electrolyte domains ($\Omega_{CP} \cup \Omega_{el}$) and tracks the ions in the electrolyte and in the PSS phase. The two sets, discussed in detail in Sec. II A, are coupled by a volumetric capacitance, C^* , usually interpreted as arising from the electrical double layer between the PEDOT filaments and the electrolyte that penetrates the polymer.²⁰

Figure 2 illustrates the cylindrical, z-axis symmetric simulation domain and defines the regions and dimensions of interest. Notice that the reference and counter electrodes are co-located at the top of the domain, for simplicity. Table I lists the default parameter values used throughout this paper, unless otherwise stated.

A. Constituting equations

The system of Poisson–Nernst–Planck equations that describes holes and ions in the PEDOT (i.e., $\vec{r} \in \Omega_{CP}$) reads¹¹

$$-\epsilon_p \nabla^2 \phi_p = pF - C_v(\phi_p - \phi_e), \quad (4)$$

$$\vec{J}_p = -FD_p \nabla p - F\mu_p p \nabla \phi_p, \quad (5)$$

$$\frac{\partial p}{\partial t} = -\frac{1}{F} \nabla \cdot \vec{J}_p, \quad (6)$$

where $p(\vec{r})$ is the molar concentration of holes, F is Faraday's constant, $\phi_p(\vec{r})$ is the potential associated to the PEDOT phase, C^* is the volumetric capacitance, $J_p(\vec{r})$ is the hole current density, D_p is the hole diffusion coefficient, and μ_p is the hole mobility.

A matching PNP set of equations is solved for ions (K^+ and Cl^- in our case) over $\Omega_{CP} \cup \Omega_{el}$,

$$-\epsilon_e \nabla^2 \phi_e = F(p + c_K - c_{Cl} - R_{CP}), \quad (7)$$

$$\vec{J}_K = -FD_K \nabla c_K - F\mu_K c_K \nabla \phi_e, \quad (8)$$

$$\vec{J}_{Cl} = +FD_{Cl} \nabla c_{Cl} - F\mu_{Cl} c_{Cl} \nabla \phi_e, \quad (9)$$

$$\frac{\partial c_K}{\partial t} = -\frac{1}{F} \nabla \cdot \vec{J}_K, \quad (10)$$

$$\frac{\partial c_{Cl}}{\partial t} = +\frac{1}{F} \nabla \cdot \vec{J}_{Cl}, \quad (11)$$

where $c_{Cl}(\vec{r})$ and $c_K(\vec{r})$ are the molar concentrations of chlorine and potassium ions, respectively, R_{CP} is the molar concentration of fixed negative charges in the PSS phase, and ϕ_e is the electrostatic potential of the PSS and electrolyte phase. Clearly, p and R_{CP} are null in the electrolyte region. Note that Eqs. (4)–(6) and (7)–(11) do not include distributed redox reactions, nor they account for overoxidation dynamics that can inactivate PEDOT chains.²³ Consistently, in Sec. III, we simulate the system in a limited voltage range $[-1.5; 0.5]$ V where overoxidation is negligible for PEDOT:PSS films immersed in aqueous electrolytes. However, overoxidation could cause a degradation of the total density of states and the hole mobility; therefore, the state of overoxidation is an important parameter even in our safe voltage range. This aspect will be discussed further in Sec. IV.

In Ref. 11, the two-phase model was developed and solved under the assumption of MB statistics. In this paper, we extend the model to account for the FD statistics. This extension requires adaptations to three main aspects: the boundary conditions, the relationship between mobility and diffusivity, and the solution scheme to achieve convergence. These aspects will be analyzed separately in Secs. II B–II D.

B. Boundary conditions

The adoption of the equilibrium Fermi–Dirac occupation function (hereafter embraced also in near-equilibrium conditions, as those under investigation) intimately changes the relation between ϕ_p and p . If we assume that the Fermi level of the PEDOT phase is perfectly aligned with that of the metal (so that the work function of the metal contact to the PEDOT:PSS layer does not affect the boundary conditions but merely shifts the vacuum level in the metal, in line with Ref. 11), we have

$$W_{AgCl} + qV_{app} = \eta|_{\partial\Omega_{elc}} + q\phi_p|_{\partial\Omega_{elc}}, \quad (12)$$

where q is the elementary charge and $\eta(\vec{r})$ is the hole chemical potential. Furthermore, the hole concentration $p(\vec{r})$ as a function of chemical potential is given by the product of the DoOS integral

and the total number of available hole sites, that is, [Eq. (3)],

$$p(\vec{r}) = p_t \cdot G(\eta(\vec{r}), E_{\text{DoS}}, \sigma). \quad (13)$$

Assuming MB statistics, and analytically working out the integral in Eq. (3), we obtain the expression of η as a function of p used in Ref. 11,

$$\eta(\vec{r}) = E_{\text{DoS}} - \frac{\sigma^2}{2k_B T} + k_B T \ln\left(\frac{p(\vec{r})}{p_t}\right), \quad (14)$$

which can be evaluated at the contact and substituted in Eq. (12) to find

$$\phi_p|_{\partial\Omega_{\text{ctc}}} = \frac{W_{\text{AgCl}}}{q} + V_{\text{app}} - \frac{1}{q} \left[E_{\text{DoS}} - \frac{\sigma^2}{2k_B T} + k_B T \ln\left(\frac{p|_{\partial\Omega_{\text{ctc}}}}{p_t}\right) \right]. \quad (15)$$

Furthermore, by assuming that the holes' charge on the PEDOT filaments locally balances the microscopic electrical double layer (EDL) charge given by the internal volumetric capacitance at the contact [i.e., assuming the net charge density at the right end side of Eq. (4) is zero], we obtain

$$p|_{\partial\Omega_{\text{ctc}}} = \frac{C_v}{F} (\phi_p - \phi_e)|_{\partial\Omega_{\text{ctc}}}. \quad (16)$$

Equations (15) and (16) were used in Ref. 11 as boundary conditions for the hole potential and the hole concentration on $\partial\Omega_{\text{ctc}}$.

Since the energy integral in Eq. (3) cannot be worked out analytically with FD statistics, we exploit Eq. (12) in a different way; namely, we cast it in the form,

$$\eta|_{\partial\Omega_{\text{ctc}}} = W_{\text{AgCl}} + qV_{\text{app}} - q\phi_p|_{\partial\Omega_{\text{ctc}}}, \quad (17)$$

and use the value $\eta|_{\partial\Omega_{\text{ctc}}}$ to compute

$$p|_{\partial\Omega_{\text{ctc}}} = p_t G(\eta|_{\partial\Omega_{\text{ctc}}}, E_{\text{DoS}}, \sigma). \quad (18)$$

In addition, Eq. (16) can be rewritten as

$$\phi_p|_{\partial\Omega_{\text{ctc}}} = \frac{F}{C^*} p|_{\partial\Omega_{\text{ctc}}} + \phi_e|_{\partial\Omega_{\text{ctc}}}, \quad (19)$$

so that Eqs. (18) and (19) can be used as implicit boundary conditions for p and ϕ_p at the contact between PEDOT and the electrode.

We complete the set of boundary conditions by imposing (1) Neumann, zero flux conditions for holes at the frontier between the polymer and the electrolyte ($\partial\Omega_{\text{interface}}$), because holes exist only in the polymer; (2) Neumann, zero flux conditions for the ions at the electrode contacting the polymer, $\partial\Omega_{\text{ctc}}$, because we neglect ions' diffusion into the metal and ion reactions with other species at the surface; and (3) Dirichlet boundary conditions for ϕ_e , c_K , and c_{Cl} on $\partial\Omega_{\text{bulk}}$. The complete set of boundary conditions

is listed below for convenience,

$$\begin{cases} p = p_t \int_{-\infty}^{\infty} \text{DoS}(E, \eta) dE, \\ \eta = W_{\text{AgCl}} + qV_{\text{app}} - q\phi_p, \\ \phi_p = \frac{F}{C^*} p + \phi_e, \quad \hat{n} \cdot \nabla \phi_e = 0, \\ \hat{n} \cdot \vec{j}_K = 0, \quad \hat{n} \cdot \vec{j}_{\text{Cl}} = 0, \end{cases} \quad \text{on } \partial\Omega_{\text{ctc}},$$

$$\hat{n} \cdot \vec{j}_p = 0, \quad \hat{n} \cdot \nabla \phi_p = 0, \quad \text{on } \partial\Omega_{\text{interface}},$$

$$c_K = c_{K,\text{bulk}}, \quad c_{\text{Cl}} = c_{\text{Cl,bulk}}, \quad \phi_e = 0 \text{ V}, \quad \text{on } \partial\Omega_{\text{bulk}}.$$

C. Relation between mobility and diffusivity

In a degenerate semiconductor, the Einstein relation expressing the diffusion coefficient as $D_p = (k_B T/q)\mu_p$ is no longer valid and should be replaced by its generalized form,^{24,25}

$$\frac{D_p}{\mu_p} = \frac{p}{q \frac{\partial p}{\partial \eta}} = \frac{p}{q \frac{\partial p}{\partial \zeta} \cdot \frac{\partial \zeta}{\partial \eta}} = \frac{k_B T}{q} \frac{p}{\frac{\partial p}{\partial \zeta}}. \quad (20)$$

Notice that the term $\partial p / \partial \zeta$ can be written in terms of C^* as in Eqs. (11) and (12) of Ref. 18 where C^* is defined as the charge variation in the PEDOT phase caused by a variation in the chemical potential, i.e.,

$$C^* = qF \frac{\partial p}{\partial \eta}. \quad (21)$$

This capacitance is implicitly included in our framework, and it models the fact that charging PEDOT entails changing its chemical potential. Under Maxwell-Boltzmann statistics, it goes as $C^* = qFp/(k_B T)$ and may become very large. On the other hand, C_v from Eq. (4) is the electrostatic capacitance between the monomers and the ions penetrating into the polymer film. The effective capacitance of the polymer is the series of C_v and C^* so that the lower the DoS, the higher impact C^* will have in determining the equivalent capacitance.

Equation (20) requires knowledge of the position-dependent chemical potential in the PEDOT. We compute $\eta(\vec{r})$ by inverting the $\text{DoS}(E)$ function,

$$\eta(\vec{r}) = G^{-1}(p(\vec{r})/p_t). \quad (22)$$

At the contact, we apply again the equilibrium assumption introduced with Eq. (17).

According to Eq. (20), the ratio between diffusivity and mobility varies in space and time according to the ratio between hole concentration and its derivative with respect to the normalized chemical potential ζ . Therefore, we may argue whether any of these two quantities (μ_p and D_p) remains constant and which one has to be varied consistently with the term $p/(\partial p / \partial \zeta)$. Given the partly unexplored nature of degenerate transport in PEDOT:PSS, in Secs. III and IV, we explore two cases where (1) mobility is constant, and diffusivity adapts to Eq. (20) (which is consistent with the choice made in TCAD tools for degenerate semiconductors, e.g.,

Ref. 26); and (2) diffusivity is constant, and mobility adapts to Eq. (20). We remark that these are limit cases and an accurate model of the field, temperature, and energy dependence of the mobility would be needed for a reliable estimation of the transport dynamics in PEDOT-PSS. Case 1 is representative of a situation where the relaxation time of the momentum is independent on the Fermi level, resulting in a constant mobility vs η .

D. Numerical implementation of the model

Lacking an analytical solution for the integral of the degenerate DoOS in Eq. (3), we used the arbitrary precision Python library *mpmath* to tabulate the integral for ten thousand values of $\zeta \in [-40, 40]$ corresponding to $\eta \in [3, 5.1]$ eV and a resolution better than 1 meV at $T = 300$ K, $E_{\text{DoS}} = 4.1$ eV. The result of the Gauss-Fermi integral is then hard-coded in COMSOL as a function of η , E_{DoS} , and σ . The derivative of the Fermi function required in Eq. (20) is also computed with *mpmath*, using a second-order accurate central differences algorithm for numerical consistency.

The numerical implementation of Eq. (22) requires special care to prevent divergent solutions. In particular, we observe that $G(\eta)$ saturates at high potential, as expected when the material approaches degenerate conditions; consequently, the inversion of $G(\eta)$ becomes challenging at high potentials. In addition, since p becomes insensitive to η , the right-hand side of Eq. (20) becomes very large in degenerate conditions. If we keep the mobility constant and compute the diffusivity according to Eq. (20), also D_p becomes large, thus hindering convergence of the PNP equations for holes. Therefore, simulations at constant mobility require smaller time steps and an implementation of the Newton method tailored for highly nonlinear problems²⁷ in time-dependent simulations.

The implementation of the AC model requires special care. While we used COMSOL Multiphysics's internal linearization procedure to obtain the AC model, we avoided using the default "harmonic perturbation" nodes. To force a small signal perturbation to the system we used the *linper()* command, given the convergence difficulties for the heavily customized multiphysics coupling that builds the two-phase model.

E. Energy diagrams

The model proposed in Sec. II was first used to gain insight into the energy alignments among the materials composing the structure. Figure 3 shows three diagrams: at equilibrium (center); with fully reduced PEDOT (left); and with fully oxidized PEDOT (right). The reference energy is the vacuum level at the reference electrode, where electrons are exchanged between the electrode and the electrolyte. We observe the following:

- The Fermi level of the ions ($E_{F,e}$, blue line) aligns to the Fermi level of the reference electrode and remains constant over space since the ion flux is null.
- Since the two phases of the PEDOT:PSS are only coupled via the volumetric capacitance C_v , there is no explicit exchange of electrons between the PEDOT phase and the electrolyte. Therefore, there are two separate Fermi levels in the polymer whose difference is equal to the bias voltage V_{app} .
- The Fermi level of the holes in the PEDOT phase ($E_{F,p}$, red line) is constant over space in DC since the hole flux is null.
- The vacuum levels (one for the PSS/electrolyte and one for the PEDOT phase) across the structure are modulated by the local electrical potentials of ions ϕ_e and holes ϕ_p , respectively.
- The small ϕ_e step at the junction between the electrolyte and the polymer is due to the different ion concentrations in the two domains.

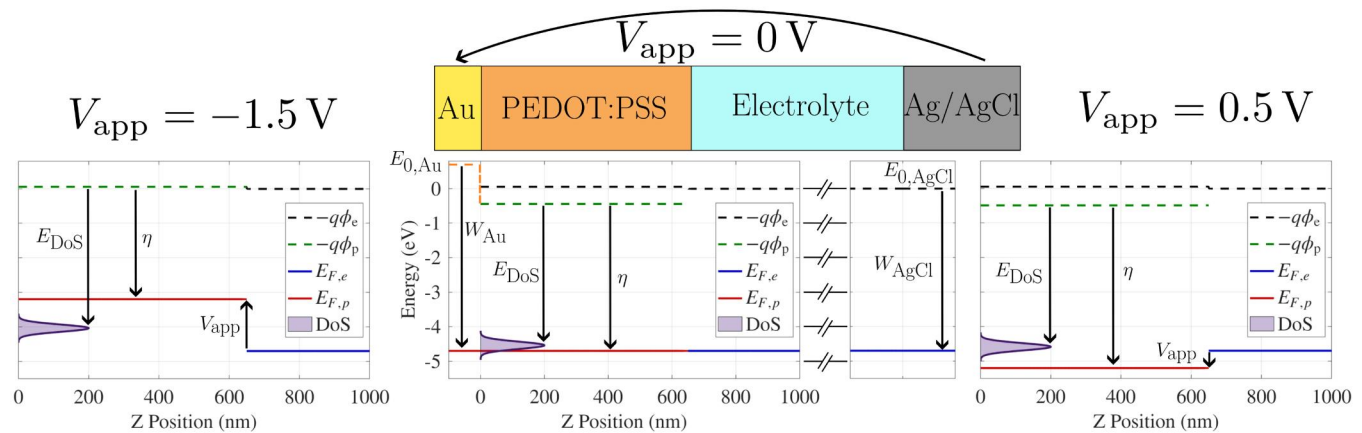


FIG. 3. Simulated energy diagram of the Au/PEDOT:PSS/electrolyte/AgCl/Ag electrode system at equilibrium, i.e., $V_{\text{app}} = 0$ V (center), $V_{\text{app}} = -1.5$ V corresponding to completely reduced PEDOT (left) and $V_{\text{app}} = 0.5$ V corresponding to completely oxidized PEDOT (right). Parameters as in Table I and $\rho_t = 166 \text{ mM} = 10^{20} \text{ cm}^{-3}$. $E_{F,e}$ and $E_{F,p}$ are the Fermi levels of the electrolyte phase and of the PEDOT phase, respectively, $E_{0,Au}$ and $E_{0,AgCl}$ are the vacuum levels of the gold and of the silver electrodes, respectively; W_{Au} and W_{AgCl} are their work functions.

- The position of the DoS centroid relative to the Fermi level in the PEDOT phase determines the steady state hole concentration. In the reduced state (left plot), the material is depleted of holes since the Fermi level lies above the DoS. The opposite happens in the fully oxidized state (right plot).

III. RESULTS ON CYCLIC VOLTAMMETRY

This section reports time-dependent cyclic voltammetry simulations computed with either FD or MB statistics over a wide parameter range. The curves in the figures refer to the second cycle to avoid the influence of the initial conditions. The current was estimated as the time variation in the spatial integral of the hole concentration across the whole layer, i.e.,

$$I(t) = F \frac{d}{dt} \int_{\Omega_{CP}} p(t, \vec{r}) dV. \quad (23)$$

A. Impacts of a low DoS for holes

We start by analyzing the combined effects of a low p_t and a FD occupation function.

The holes' DoS, hence p_b , can be mildly modified by treatments that alter the PSS content of the film such as the addition of

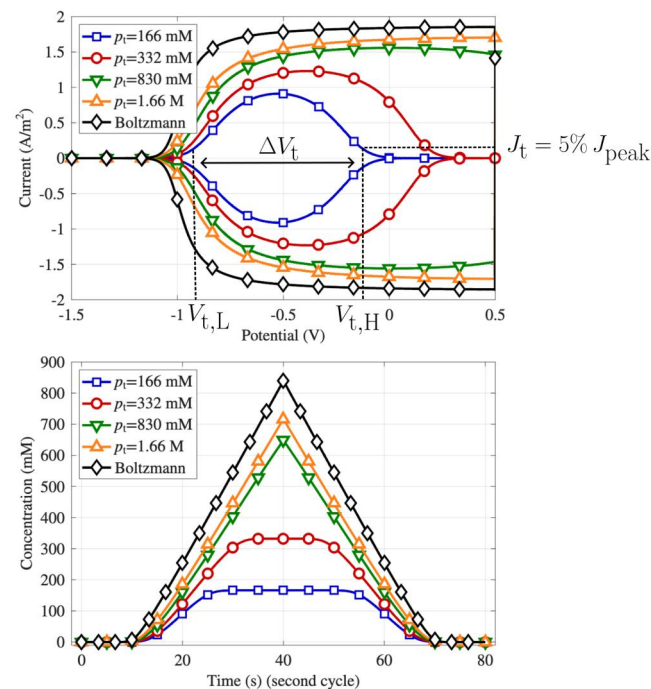


FIG. 4. Top: Simulated second cycle of the voltammetry curve using the MB ($p_t = 1.66$ M, black curve) and the FD occupation function for a few p_t values. Bottom: Corresponding hole concentration at the boundary between the PEDOT and the gold electrode. The molar concentrations correspond to $p_t = 10^{20}$, 2×10^{20} , 5×10^{20} , 10^{21} cm^{-3} . Scan rate = 50 mV/s.

sulfuric acid.²⁸ Indeed, sulfuric acid or other strong acids²⁹ will dissolve part of the excess PSS and thus reduce the overall volume of the film, which leads to increased density of the hole states in the overall PEDOT:PSS film. Notice that these treatments only affect the PSS content and the microscopic morphology,³⁰ so they do not alter the DoS of PEDOT itself. Overoxidation at large voltages was also shown to dynamically change the PEDOT conductivity²³ and could be a possible cause of degraded p_t values. In our simulations, the minimum density of hole sites is chosen as $p_t = 10^{20} \text{ cm}^{-3}$ according to Ref. 31. A reasonable upper bound of 10^{23} cm^{-3} (consistent with Ref. 31), can be inferred from the packing density of organic semiconductors. Practically, $p_t \leq 10^{21} \text{ cm}^{-3}$ was chosen, since the CV curves do not change much above this value.

Figure 4 (top) shows a few CV curves computed for different p_t values with degenerate transport [Eq. (20)] and constant mobility. When p_t is large (e.g., $10^{21} \text{ cm}^{-3} = 1.66$ M), the conjugated polymer operates far from degenerate conditions during the whole cycle. Nonetheless, the maximum current and the voltage aperture of the CV eye are appreciably overestimated if MB statistics is assumed (compare black and orange curves). p_t has modest effects on the curves computed with MB statistics; so, only the case $p_t = 10^{21} \text{ cm}^{-3}$ is shown. In case of FD statistics, instead, when p_t decreases, the rise of the current moves to larger voltages (≈ -1 V) and shows a smaller slew rate. Furthermore, the transition from positive to negative current becomes gradual, and we observe a reduction in the capacitive eye width (ΔV_t in the top plot). The saturation of the hole concentration, arising from the filling of all available states (Fig. 4, bottom panel), leads to a current reduction relative to non-degenerate conditions.

As it will be shown in Fig. 8, the choice among Einstein, non-degenerate transport, or degenerate transport with either constant mobility or constant diffusivity, does not affect the simulated cyclic voltammogram. Therefore, the sensitivity of the voltage where eye closure occurs ($V_{t,H} \approx 0$ V in Fig. 4) to the p_t value can be exploited to quantitatively monitor the degeneracy of the polymer (or the degree of overoxidation, should this be identified as the cause of p_t reduction).

Figure 5 shows the relation between p_t , $V_{t,H}$, and ΔV_t for the same system as in Fig. 4. We observe that the relations are all quite linear except for the point of minimum p_t , where the mobility is extremely low.³¹

The absolute $V_{t,H}$ values shift rigidly with the energy centroid of the DoS (E_{DoS}) as shown in Fig. 6. Therefore, while the value of E_{DoS} can be tuned to match cyclic voltammetry or ultraviolet photoemission spectroscopy or x-ray photoemission spectroscopy (UPS/XPS), the measured $V_{t,H}$ and ΔV_t sensitivity to p_t reported in Fig. 5 can be taken as a monitor of the PEDOT:PSS density of states.

B. Impacts of a low PEDOT to PSS ratio

PEDOT and PSS chains can arrange differently at a microscopic level depending on the fabrication conditions. Pristine PEDOT:PSS exhibits a high phase separation, with PEDOT-rich and PSS-rich grains in its structure. With solvent treatment³⁰ and proper annealing, the microscopic structure can be rearranged so

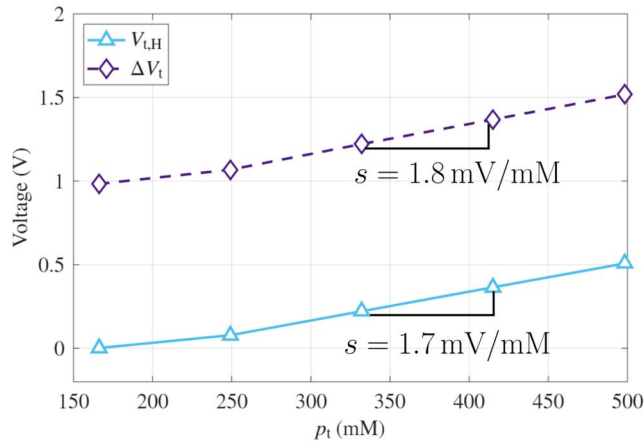


FIG. 5. Relationship between p_t and the horizontal aperture of the CV curves, quantified via the upper and lower voltages of 5% peak current and the corresponding aperture widths as indicated in Fig. 4. $p_t = 1 - 3 \times 10^{20}$ in 0.5×10^{20} steps. Scan rate = 50 mV/s.

that the PEDOT-rich regions are better connected and form more efficient charge conduction pathways.

Therefore, a lower value of μ_p is expected for pristine PEDOT:PSS, i.e., when the PEDOT grains are far apart from one another, and the fractional PSS content is large. Indeed, the overall mobility will depend on the farthest distance that holes must hop between two points of the film. Low mobility has been reported experimentally in PEDOT:PSS films with high PSS content and in films that were not treated with acid.²⁸

To investigate how drift-diffusion transport is degraded by a low PEDOT to PSS ratio or by inadequate film treatments, we

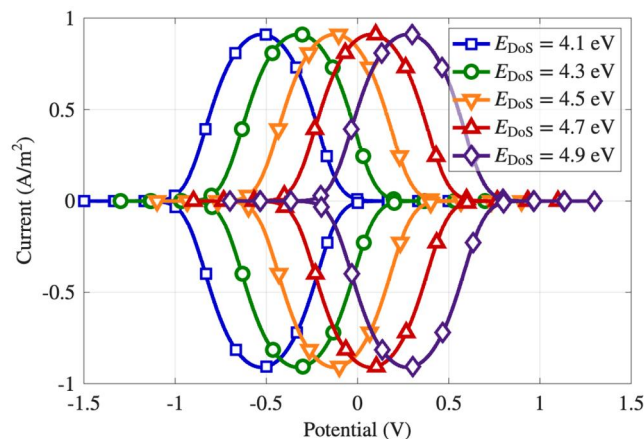


FIG. 6. Simulated second cycle of the voltammetry curves using the FD occupation function for different centroid of the Gaussian DoS (E_{DoS}). Scan rate = 50 mV/s.

reduce either the mobility or the diffusivity by a scaling factor α as per the relations,

$$\mu_p = \alpha \cdot \mu_{p0} \quad \text{or} \quad D_p = \alpha \cdot D_{p0}, \quad (24)$$

where μ_{p0} and D_{p0} are given in Table I. If μ_p (D_p) is chosen according to Eq. (24), then D_p (respectively, μ_p) is computed according to Eq. (20) and vice versa, thus respecting the constraints set by the material's degeneracy. In both cases, the concentration of PSS counterions R_{CP} is kept at the constant value of Table I.

The effect of degrading the transport by changing α from 10^{-6} to 1 interval is shown in Fig. 7. Plots in the top (bottom) panel assume a fixed mobility (diffusivity) with the scaling factors specified in the legend.

The curves in panels Figs. 7(a), 7(b), 7(e), and 7(f) show that independently of the chosen implementation of generalized Einstein transport, the rise and the fall of the current during the CV cycle and the rise and fall of the hole concentration are delayed when α is decreased. This results in remarkable distortions of the cyclic voltammogram, and asymmetries between the rise and fall edges are consistent with the reduced drift-diffusion current. Notably, the shape of the CV curve for constant diffusivity and $\alpha = 10^{-6}$ [Fig. 7(e)] resembles that of a purely Faradaic electrode. Furthermore, the peak current varies linearly with the square root of the scan rate [orange curve in Fig. 7(g)], as expected for Faradaic electrodes.³²

However, our simulations do not implement redox processes at the electrode but only the intrinsic doping mechanism of PEDOT:PSS involving the EDOT monomer and the K^+ cation; that is, $\text{EDOT}^+ + e^- + K^+ \rightleftharpoons \text{EDOT}^0 \cdot K^+$. In fact, this reaction simply means that when an electron provided by the electrode contact recombines with a localized hole in the EDOT monomer, the free cation (in this case K^+ coming from the electrolyte) balances the PSS negative charge. The electrostatics of this doping mechanism is included in the model [Eqs. (4) and (7)] so that the conjugated polymer is accumulated or depleted of holes depending on the concentration of intercalated ions, but no real chemical reaction supporting DC currents at the electrode is implemented.

These results imply that in the presence of a low PEDOT:PSS content ratio, and/or a low amount of PEDOT with low hole diffusivity, we could misinterpret the CV shape as due to Faradaic processes.

Instead, if p_t is large enough to sustain the hole current during the whole voltage scan without running into degenerate conditions [Figs. 7(d) and 7(h)], then a peak appears in the cyclic voltammogram when reducing the mobility or the diffusivity. However, the current remains mostly capacitive, similarly to the experiments and the simulations with MB statistics and Einstein relation reported in Ref. 11.

Therefore, the simulations in Fig. 7 suggest that when holes suffer from a degraded mobility or diffusivity, $\alpha \leq 10^{-4}$ (see Fig. 8), the adoption of a generalized Einstein relation in the drift-diffusion transport model is quite relevant.

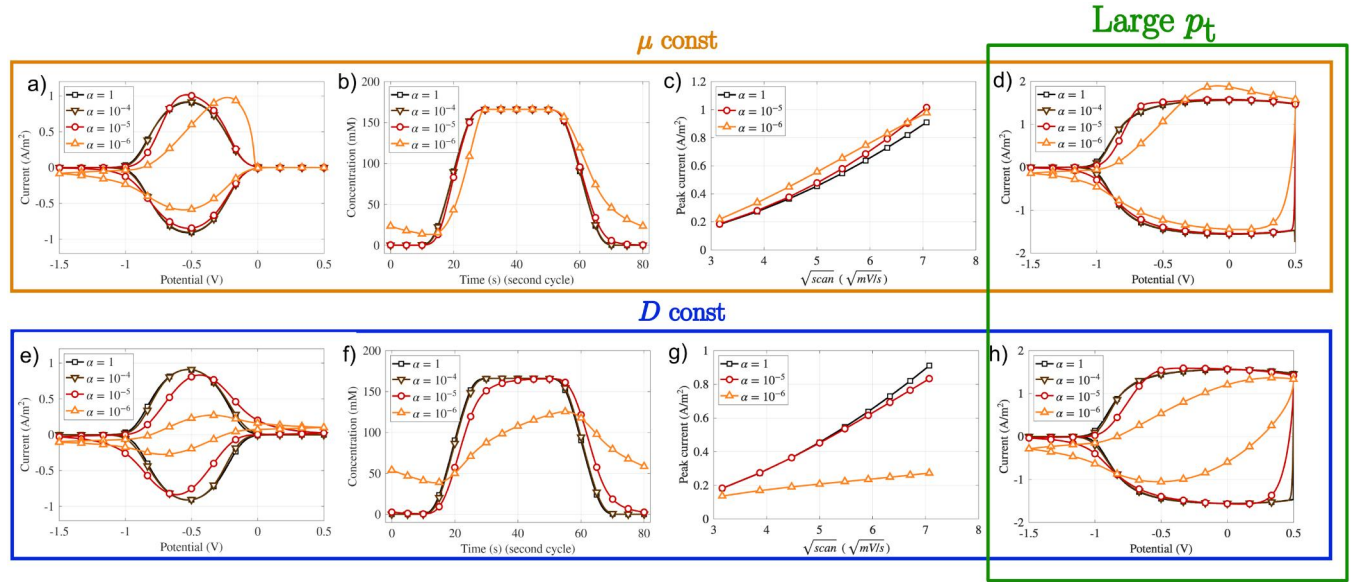


FIG. 7. Simulations with constant μ_p (top panel) and constant D_p (bottom panel) versions of the generalized Einstein relation [Eq. (20)]. First column, (a) and (e): CV curves obtained with FD occupation and $p_t = 10^{20} \text{ cm}^{-3} = 166 \text{ mM}$ for varying α , mimicking varying intersite hopping distance. Second column, (b) and (f) corresponding hole concentration at the gold contact. Third column, (c) and (g) peak value of the current for a varying scan rate. Fourth column, (d) and (h) CV curves obtained with FD occupation and $p_t = 5 \times 10^{20} \text{ cm}^{-3} = 830 \text{ mM}$.

IV. RESULTS ON ELECTROCHEMICAL IMPEDANCE SPECTROSCOPY

Besides cyclic voltammetry, the physical characterization of conjugated polymers often relies on electrochemical impedance spectroscopy (EIS). It is, therefore, interesting to examine the impact of FD statistics and degenerate transport due to a limited hole DoS and to a deteriorated mobility or diffusivity. Consistently, we use the linearized, time-harmonic (AC) formulation of the model equations (4)–(6) and (7)–(11) to analyze the EIS response of the polymer; this was achieved with the default COMSOL linearization procedure.²⁷ Simulations are performed at a bias $V_{app} = 0 \text{ V}$ and over a wide frequency range from 0.01 Hz to 1 MHz. Analogously to Eq. (23), the current through the layer was

estimated as

$$I_{AC}(\omega) = j\omega F \int_{\Omega_{CP}} p_{AC}(\omega) dV. \quad (25)$$

Figure 9 shows typical simulated Cole–Cole plots of the system impedance for varying p_t and α in the generalized Einstein transport scenarios examined in Sec. III. All curves exhibit a vertical branch at low frequency, an expected signature of capacitive behavior constrained by a series resistance. When comparing Figs. 9(a) and 9(b), we observe a shift of this series resistance in the constant diffusivity case, which does not appear in the constant mobility case. The reason is that, for the chosen steady state bias point $V_{app} = 0 \text{ V}$, the hole concentration is saturated at its maximum value. Therefore, if the mobility is constant, the conductivity $qp\mu_p$ is also constant; on the other hand, if the diffusivity is constant, then μ_p changes according to the generalized Einstein relation and the resistance changes as well. If we enforce a reduced mobility or diffusivity by decreasing α [Figs. 9(c) and 9(d), respectively], the Warburg-like component of the impedance, which is responsible for the 45° slope of the curves,³² becomes more prominent. Indeed, a smaller α factor induces a larger series resistance; consequently, a larger portion of the curve with 45° slope becomes visible. The markedly different features of $\text{Re}(Z)$ and $\text{Im}(Z)$ suggest the possibility of gaining insights into the physical properties of the PEDOT:PSS by scrutiny of the CV and EIS curves.

Figure 10 shows the same data as in Figs. 9(a) and 9(c) but in the form of Bode plots.

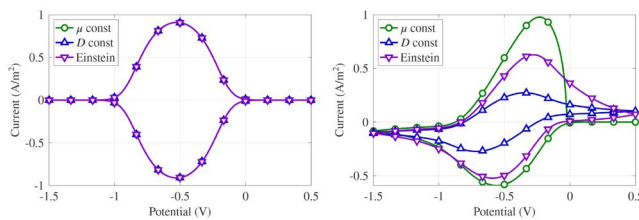


FIG. 8. CV curves obtained with FD occupation and $p_t = 10^{20} \text{ cm}^{-3} = 166 \text{ mM}$, for constant μ_p , constant D_p , and Einstein relation with $\alpha = 1$ (left) and $\alpha = 10^{-6}$ (right). Scan rate = 50 mV/s.

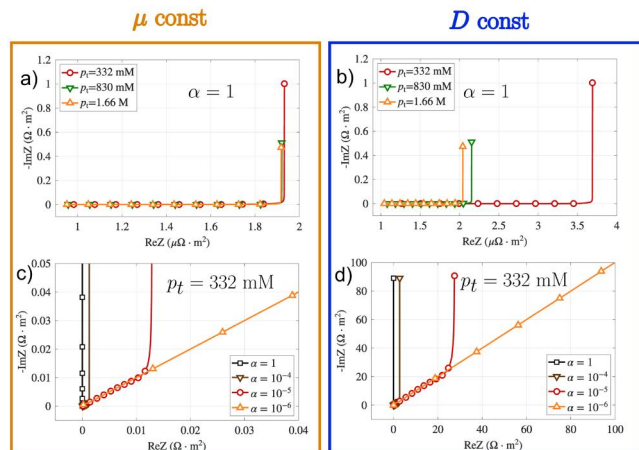


FIG. 9. Cole-Cole plot of the electrochemical impedance vs frequency with FD occupation for generalized Einstein transport and varying p_t [plots (a) and (b)] and varying diffusivity or mobility degradation factor α . Plots (a) and (c) are for different p_t and α , respectively, at constant mobility. Plots (b) and (d) same as (a) and (c) but at constant diffusivity.

All plots show a capacitive behavior at low frequency (amplitude decreasing with -20 dB/dec combined with a -90° phase). Figures 10(a) and 10(b) show that the mobility degradation introduces a 45° phase plateau at mid-to-high frequency (100 Hz–1 MHz), which is indicative of increasing diffusion-limited transport processes. Notice that the transition frequency from purely capacitive to diffusive behavior shifts down with increasing mobility degradation.

Figures 10(b) and 10(d) show that reducing the DoS reduces the effective total capacitance of the material; indeed, the

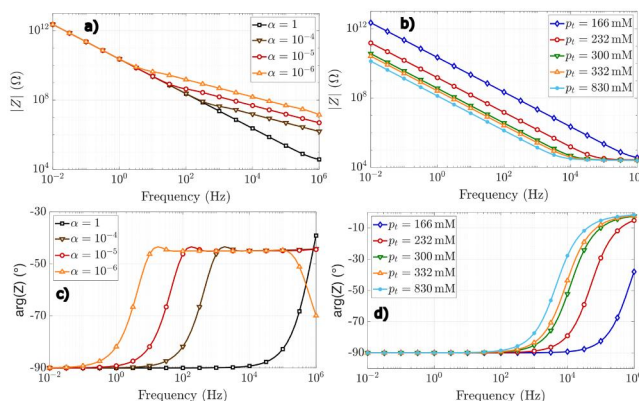


FIG. 10. Bode plots with FD occupation and generalized Einstein transport with constant μ_p . Panels (a) and (b) show the magnitude, and panels (c) and (d) show the phase of the impedance. Panels (a) and (c) have fixed total density of states and varying mobility degradation; the vice versa holds for panels (b) and (d).

impedance below the cutoff frequency (around 10 kHz) increases, i.e., lower capacitance, when reducing p_t . This is due to the fact that with lower p_t and for the same bias point, we obtain a higher degree of degeneracy; therefore, the AC charge that can be injected in the PEDOT:PSS is lower due to the filling of the DoS, which in turn results in a lower effective capacitance. For $p_t > 830$ mM, the low frequency impedance does not change appreciably (not shown).

V. CONCLUSIONS

Degeneracy effects can have an important role on the transport properties of holes in conjugated polymers. This paper introduces a new, improved model to reproduce the most common characterization tests on PEDOT:PSS samples (CV and EIS). A detailed analysis of CV and EIS curves showed that a closure of the voltage and current amplitude of the CV eye can be expected in PEDOT:PSS samples featuring low hole DoS. A relation between the total density of hole states and the shape of the capacitive eye of the CV curves. Our model can be useful to interpret the experiments more accurately.

The study of cyclic voltammograms with a varying content of PSS and the presence of the degeneracy led us to identify conditions where degeneracy and deterioration of hole transport could be misinterpreted as the insurgence of a Faradaic behavior at the electrode featuring the expected linear trend vs the square root of the scan rate, in spite of being of totally different origin.

Although the model does not include redox reactions and electrolyte discharge (that could possibly mask the closure of the eye at high applied voltages), it can still be useful to extricate the complex consequences of hole degeneracy and degraded hole transport on CV and EIS curves and to extract more accurate physical parameter values from measurements.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jacopo Nicolini: Conceptualization (equal); Data curation (equal); Investigation (equal); Software (lead); Writing – original draft (equal); Writing – review & editing (equal). **Pierpaolo Palestri:** Conceptualization (equal); Investigation (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal). **Luca Selmi:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹A. Benoudjit, M. M. Bader, and W. W. A. Wan Salim, *Sens. Bio-Sensing Res.* **17**, 18 (2018).
- ²J. Bobacka, A. Lewenstam, and A. Ivaska, *J. Electroanal. Chem.* **489**, 17 (2000).
- ³N. Gao, J. Yu, Q. Tian, J. Shi, M. Zhang, S. Chen, and L. Zang, *Chemosensors* **9**, 79 (2021).
- ⁴I. Gualandi, M. Marzocchi, E. Scavetta, M. Calieni, A. Bonfiglio, and B. Fraboni, *J. Mater. Chem. B* **3**, 6753 (2015).
- ⁵M. Heydari Gharahcheshmeh, M. T. Robinson, E. F. Gleason, and K. K. Gleason, *Adv. Funct. Mater.* **31**, 2008712 (2021).
- ⁶A. Savva, C. Cendra, A. Giugni, B. Torre, J. Surgailis, D. Ohayon, A. Giovannitti, I. McCulloch, E. Di Fabrizio, A. Salleo, J. Rivnay, and S. Inal, *Chem. Mater.* **31**, 927 (2019).
- ⁷H. Bässler, *Phys. Status Solidi B* **175**, 15 (1993).
- ⁸R. Coehoorn, W. F. Pasveer, P. A. Bobbert, and M. A. J. Michels, *Phys. Rev. B* **72**, 155206 (2005).
- ⁹T. Stöcker, A. Köhler, and R. Moos, *J. Polym. Sci., Part B* **50**, 976 (2012).
- ¹⁰S. Bitton, P. Alarcon-Espejo, A. F. Paterson, and N. Tessler, *J. Appl. Phys.* **136**, 125501 (2024).
- ¹¹K. Tybrandt, I. V. Zozoulenko, and M. Berggren, *Sci. Adv.* **3**, eaao3659 (2017).
- ¹²I. Zozoulenko, A. Singh, S. K. Singh, V. Gueskine, X. Crispin, and M. Berggren, *ACS Appl. Polym. Mater.* **1**, 83 (2019).
- ¹³J. L. Bredas and G. B. Street, *Acc. Chem. Res.* **18**, 309 (1985).
- ¹⁴S. Ihnatsenka, X. Crispin, and I. V. Zozoulenko, *Phys. Rev. B* **92**, 035201 (2015).
- ¹⁵G. Kim and K. P. Pipe, *Phys. Rev. B* **86**, 085208 (2012).
- ¹⁶M. Javadi, A. Mazaheri, H. Torbatiyan, and Y. Abdi, *Phys. Rev. Appl.* **12**, 034002 (2019).
- ¹⁷M. Oehzelt, N. Koch, and G. Heimel, *Nat. Commun.* **5**, 4174 (2014).
- ¹⁸S. T. Keene, A. Rao, and G. G. Malliaras, *Sci. Adv.* **9**, eadi3536 (2023).
- ¹⁹T. J. Whitcher, W. S. Wong, A. N. Talik, K. L. Woon, N. Chanlek, H. Nakajima, T. Saisopa, and P. Songsiririthigul, *J. Phys. D: Appl. Phys.* **49**, 325106 (2016).
- ²⁰A. V. Volkov, K. Wijeratne, E. Mitraka, U. Ail, D. Zhao, K. Tybrandt, J. W. Andreasen, M. Berggren, X. Crispin, and I. V. Zozoulenko, *Adv. Funct. Mater.* **27**, 1700329 (2017).
- ²¹L. Yuan-Hui and S. Gregory, *Geochim. Cosmochim. Acta* **38**, 703 (1974).
- ²²J. Ouyang, C.-W. Chu, F.-C. Chen, Q. Xu, and Y. Yang, *Adv. Funct. Mater.* **15**, 203 (2005).
- ²³P. Tehrani, A. Kancirzewska, X. Crispin, N. Robinson, M. Fahlman, and M. Berggren, *Solid State Ionics* **177**, 3521 (2007).
- ²⁴Y. Roichman and N. Tessler, *Appl. Phys. Lett.* **80**, 1948 (2002).
- ²⁵N. W. Ashcroft and N. D. Mermin, *Solid State Physics* (Cengage, Boston, MA, 2023).
- ²⁶Synopsys, Inc., *Sentaurus™ Device User Guide*, Synopsys, Inc., Mountain View, CA (2023), version U-2022.12.
- ²⁷COMSOL AB, *COMSOL Multiphysics® Reference Manual*, COMSOL AB, Stockholm (2024), version 6.3.
- ²⁸E. Hosseini, V. Ozhukil Kollath, and K. Karan, *J. Mater. Chem. C* **8**, 3982 (2020).
- ²⁹H. Yousefian, S. A. Hashemi, A. Babaei-Ghazvini, B. Acharya, A. Ghaffarkhah, and M. Arjmand, *Mater. Adv.* **5**, 4699 (2024).
- ³⁰M. Modarresi and I. Zozoulenko, *Phys. Chem. Chem. Phys.* **24**, 22073 (2022).
- ³¹D. Lee, J. Zhou, G. Chen, and Y. Shao-Horn, *Adv. Electron. Mater.* **5**, 1800624 (2019).
- ³²A. J. Bard, L. R. Faulkner, and H. S. White, *Electrochemical Methods: Fundamentals and Applications* (John Wiley & Sons, 2022).